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When Your Model Stops Working: Anytime-Valid Calibration Monitoring

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Abstract: Practitioners monitoring deployed probabilistic models face a fundamental trap: any fixed-sample test applied repeatedly over an unbounded stream will eventually raise a false alarm, even when the model is perfectly stable. We present PITMonitor, an *anytime-valid* calibration monitor built on the e-process framework. By testing the exchangeability of Probability Integral Transforms (PITs), PITMonitor guarantees $\mathbb{P}(\text{ever alarm} \mid H_0) \leq \alpha$ for all time, with no pre-specified monitoring horizon. We prove Type I error control via Ville's inequality and evaluate PITMonitor on three drift scenarios from `river`'s FriedmanDrift dataset, comparing against the seven included detectors. PITMonitor achieves competitive detection rates on global drift scenarios, with substantially longer detection delays on local expanding drift, while being the only calibration-specific monitor in the comparison to provide both an anytime-valid false alarm guarantee and Bayesian changepoint localization. Code is available at `https://github.com/tristan-farran/pitmon`.

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1. Introduction

Probabilistic models deployed in production face a fundamental challenge: the world changes. Across domains, from medicine to finance, models encounter regime shifts and concept drift which can cause calibration to degrade drastically with consequential effects downstream. Practitioners must therefore monitor their models continuously, seeking to answer two questions: *has* the world changed, and if so, *when*? In practice, this monitoring often relies on ad-hoc procedures such as periodic recalibration schedules, rolling-window hypothesis tests, threshold-based alerts on summary metrics, or manual inspection of residuals.

These approaches suffer from a fundamental statistical problem: *they do not control the false alarm rate over continuous monitoring*. A practitioner who checks calibration daily with a $p < 0.05$ threshold will, over a year of monitoring, almost certainly observe spurious alarms even if the model remains stable. Classical hypothesis tests assume a fixed sample size determined before seeing data; continuous monitoring violates this assumption.

More principled alternatives are provided by online drift detectors, such as those implemented in the `river` library (Montiel et al., 2021). Classical detectors including DDM, EDDM, and KSWIN are lightweight, easy to deploy, and often effective at detecting abrupt changes, but are typically based on heuristic thresholds or fixed-sample statistical arguments and do not provide explicit long-run false alarm guarantees under continuous monitoring, nor changepoint localization.

ADWIN (Bifet and Gavaldà, 2007) improves on fixed-window methods by adapting its window size to detected changes, bounding the false alarm probability *within a single window*.

However, this per-window guarantee does not directly imply stream-level control over an unbounded monitoring horizon because the number of tests grows unboundedly with the monitoring horizon. Additionally, ADWIN operates on generic signals such as error rates or residuals, and is not designed to detect the distributional shifts in PIT values that characterize calibration drift. For example, a model that becomes systematically overconfident without affecting accuracy would alter the PIT distribution but leave the mean behaviour of residuals unchanged and may therefore be missed by ADWIN.

We propose PITMonitor, an anytime-valid calibration monitoring method with five key properties:

1. **Anytime-valid false alarm control:** we prove that $\mathbb{P}(\text{ever alarm} \mid H_0) \leq \alpha$ for all time, without requiring a pre-specified monitoring horizon or stopping rule.
2. **Change detection and localization without static-error alarms:** PITMonitor detects and locates *changes* in the PIT process. A model that is consistently miscalibrated but stable will not trigger alarms, while changes in calibration can.[1]
3. **No baseline period required:** unlike methods requiring a "clean" reference distribution, PITMonitor works from the first observation.
4. **Practical efficiency:** the method requires $O(1)$ computation per update of the mixture e-process, plus $O(\log n)$ for rank maintenance via a sorted data structure, with $O(n)$ total memory.
5. **Distribution-free:** PITMonitor does not require pre- or post-change distributions.

2. Related Work

Calibration Assessment

Classical calibration metrics include expected calibration error (Guo et al., 2017), reliability diagrams (DeGroot and Fienberg, 1983), and proper scoring rules (Gneiting and Raftery, 2007). Modern neural networks can improve classification accuracy while becoming less calibrated (Guo et al., 2017), which motivates monitoring calibration online rather than only after deployment. PITs have been used for static forecast evaluation in econometrics (Diebold, Gunther and Tay, 1998) and weather prediction (Gneiting and Katzfuss, 2014). However, both these metric-based diagnostics and PIT-based evaluations are typically point-in-time and do not address sequential monitoring with false alarm control.

Distribution Shift Detection

Methods for detecting covariate shift include two-sample tests (Rabanser, Günnemann and Lipton, 2019), domain classifiers (Lipton, Wang and Smola, 2018), and conformal methods for prediction under label shift (Podkopaev and Ramdas, 2021). These primarily address changes in the input or label distributions, while our work focuses on the *output* side: identifying when predicted probabilities no longer match outcome frequencies.

[1] In many domains some amount of miscalibration is inevitable, but degradation remains a pressing concern.

Sequential Calibration Testing

Arnold, Henzi and Ziegel (2023) proposed e-values for testing forecast calibration, focusing on whether PITs are Uniform$(0, 1)$. PITMonitor differs in two ways. First, we test exchangeability rather than uniformity and thus alarm only on changes in calibration, the more natural null where some level of miscalibration is inevitable. Second, we attempt to locate *when* calibration changed, rather than only establishing whether the model is miscalibrated at a point in time.

E-values

The e-value framework has seen rapid development (Vovk and Wang, 2021; Ramdas et al., 2023; Grünwald, de Heide and Koolen, 2024; Shin, Ramdas and Rinaldo, 2024). PITMonitor builds on this framework by constructing a mixture e-process to test exchangeability of PIT values, inheriting the anytime-valid false alarm guarantees this literature provides (Theorem 1).

Changepoint Detection

Classical methods include CUSUM (Page, 1954) and Shiryaev-Roberts procedures (Shiryaev, 1963; Pollak, 1985). These assume known pre-change and post-change distributions, while our approach provides nonparametric changepoint detection with false alarm guarantees.

3. Background

3.1. Probability Integral Transforms

A probabilistic model outputting a predicted cumulative distribution function $\hat{F}$ over outcomes is *calibrated* if these predictions match reality: among all predictions where $\hat{F}(y) = p$, the outcome $Y \leq y$ should occur approximately $(100 \times p)\%$ of the time.

The *probability integral transform* (PIT) $U = \hat{F}(y)$, provides a universal tool for assessing calibration, as it satisfies $U \sim \text{Uniform}(0, 1)$ whenever $\hat{F}$ is the true distribution of Y (Dawid, 1984). Therefore, non-uniformity of the PIT distribution indicates miscalibration, and changes in the PIT distribution indicate changes in calibration.

3.2. Exchangeability

A sequence $(X_1, X_2, \ldots)$ is *exchangeable* if its joint distribution is invariant to finite permutations. Exchangeability is weaker than independence: i.i.d. sequences are exchangeable, but exchangeable sequences need not be independent (de Finetti, 1937).

Remark 1 (Stable Miscalibration Preserves Exchangeability). Under stable miscalibration, PITs are i.i.d. from some fixed, non-uniform distribution; since i.i.d. sequences are exchangeable, the PIT process remains exchangeable.

This observation is central to PITMonitor's design:

- **Perfect calibration:** PITs are i.i.d. Uniform$(0, 1) \Rightarrow$ exchangeable
- **Stable miscalibration:** PITs are i.i.d. from a non-uniform distribution $\Rightarrow$ exchangeable

- **Changing calibration:** the PIT distribution changes at some time $\tau \Rightarrow$ not exchangeable

Thus, by testing exchangeability directly, we aim to alarm only on *changes* in calibration.[2]

3.3. Conformal P-values

To test exchangeability sequentially we employ *conformal p-values from ranks* (Vovk, Gammerman and Shafer, 2005) to yield a fully non-parametric construction.

Given observations $U_1, \ldots, U_t$, define the rank of U_t:

$$R_t = \#\{s \leq t : U_s \leq U_t\} \tag{1}$$

Proposition 1 (Rank Uniformity under Exchangeability)**.** *If $(U_1, \ldots, U_t)$ is exchangeable, then the rank R_t is uniformly distributed on $\{1, \ldots, t\}$.*

Proof. Since PITs arise from a continuous predictive CDF, ties occur with probability zero, so the rank R_t is almost surely well-defined. By exchangeability, $(U_1, \ldots, U_t)$ is equally likely to be in any of the $t!$ orderings. For any fixed rank $r \in \{1, \ldots, t\}$, exactly $(t-1)!$ of these orderings place U_t in position r. Therefore, $\mathbb{P}(R_t = r) = (t-1)!/t! = 1/t$. □

In practice, deployed models may output discrete, rounded, or quantized probabilities, producing ties. In this case, randomized tie-breaking (assigning tied observations a uniformly random order among themselves) preserves the uniform rank property.

While Proposition 1 establishes that R_t is discrete uniform on $\{1, \ldots, t\}$, our density-based e-value construction requires $p_t \sim \text{Uniform}(0, 1)$ exactly. Adding a uniform jitter converts the discrete rank to an exactly continuous uniform p-value:[3]

$$p_t = \frac{R_t - 1 + V_t}{t}, \quad V_t \sim \text{Uniform}(0, 1) \tag{2}$$

Under H_0 (exchangeability), these p-values are $\text{Uniform}(0, 1)$. After a changepoint, exchangeability breaks and new PITs systematically rank higher or lower than pre-change PITs.

3.4. E-values

An *e-value* is a non-negative random variable e satisfying $\mathbb{E}[e] \leq 1$ under the null hypothesis. By Markov's inequality, $\mathbb{P}(e \geq 1/\alpha) \leq \alpha$, so rejecting when $e \geq 1/\alpha$ defines a valid level-α test, with power whenever $\mathbb{E}[e] > 1$ under the alternative.

The density-based construction in Section 4.1 achieves this adaptively: when p-values concentrate in certain bins due to non-exchangeability, the histogram places more mass there, typically yielding $\mathbb{E}[e] > 1$ without requiring a parametric specification of the alternative.

A key property for sequential monitoring is that e-values can be composed multiplicatively while maintaining validity under the null (Ramdas et al., 2023). A cumulative e-process employs this property to accumulate evidence from time τ onwards:

$$M_t^{(\tau)} = \prod_{s=\tau}^{t} e_s, \quad M_t^{(\tau)} = M_{t-1}^{(\tau)} \cdot e_t \tag{3}$$

[2] Temporal dependence and non-stationarity in the PIT stream can also violate exchangeability, so the theoretical guarantees which hold for the idealized case may not extend to more complex scenarios. See the Limitations section.

[3] It should be noted that the jitter V_t introduces an additional source of randomness beyond the data, and thus PITMonitor's output is stochastic given a fixed stream, though the additional variability is negligible as it is $O(1/t)$.

Taking conditional expectations given past observations:

$$\mathbb{E}[M_t^{(\tau)} \mid \mathcal{F}_{t-1}] = M_{t-1}^{(\tau)} \cdot \mathbb{E}[e_t \mid \mathcal{F}_{t-1}] \leq M_{t-1}^{(\tau)} \tag{4}$$

Thus each $(M_t^{(\tau)})_{t \geq \tau}$ is a non-negative supermartingale under H_0, making it unlikely to reach the threshold by chance. Under the alternative, $\mathbb{E}[e] > 1$ causes the product to grow exponentially in expectation, with larger distributional shifts producing larger e-values and faster growth towards the threshold.

4. Method

4.1. E-values via Density Betting

We construct e-values from conformal p-values using the betting framework of Ramdas et al. (2023). At time t, before observing p_t, we choose a density function $\hat{f}_t$ with unit cost under the null, i.e., $\int_0^1 \hat{f}_t(p)\, dp = 1$. Interpreting this as a wager against H_0, we pay one unit up front and receive return $\hat{f}_t(p_t)$ after p_t is revealed, giving the one-step e-value $e_t = \hat{f}_t(p_t)$.

Proposition 2 (Density Betting Yields Valid E-values)**.** *Let $\hat{f} : [0,1] \to [0,\infty)$ be a density function satisfying $\int_0^1 \hat{f}(p)\, dp = 1$. If $p \sim \text{Uniform}(0,1)$, then $e = \hat{f}(p)$ satisfies $\mathbb{E}[e] = 1$.*

Proof. Since $p \sim \text{Uniform}(0,1)$, $\mathbb{E}[e] = \mathbb{E}[\hat{f}(p)] = \int_0^1 \hat{f}(p)\, dp = 1$. □

PITMonitor uses a plug-in histogram density estimator for betting on conformal p-values (for the broader plug-in martingale lineage, see Fedorova et al. (2012)), which places mass proportional to where past p-values concentrated and requires no specification of an alternative:

$$\hat{f}(p) = B \cdot \frac{c_b}{\sum_j c_j} \quad \text{for } p \in \text{bin } b \tag{5}$$

where c_b counts past p-values in bin b and B is the number of bins. The histogram is initialized with pseudocounts $c_b = 1$ for all b, which ensures $\hat{f}$ is a valid density from the first observation and prevents zero-count bins from generating infinite or zero e-values during early monitoring. We update the histogram *after* computing e_t, ensuring $\hat{f}$ is predictable, as required for the supermartingale property of the e-process.

By adapting our density to observed concentration patterns, we automatically bet in the right direction. Under exchangeability, p-values scatter uniformly, and $\mathbb{E}[e_t] = 1$. If exchangeability breaks, p-values cluster, the histogram learns these concentration patterns, and we achieve $\mathbb{E}[e] > 1$ whenever deviations are sufficiently persistent.

Proposition 3 (Consistency)**.**

1. *If $p_t \sim \text{Uniform}(0,1)$, then e_t is a valid one-step e-value s.t. $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] \leq 1$.*
2. *If p-values are i.i.d. non-uniform after a changepoint, then*

$$\lim_{t \to \infty} \mathbb{E}[e_t \mid \mathcal{F}_{t-1}] > 1 \quad a.s.$$

Proof. Part (1) follows from Proposition 2 since $\int_0^1 \hat{f}(p)\, dp = \sum_{b=1}^{B} \left(B \frac{c_b}{\sum_j c_j} \right) \frac{1}{B} = 1$.

For part (2), let $N_{t-1,b}$ be the number of post-change p-values in bin b up to time $t-1$, b_t be the bin containing p_t, and $\theta_b := \mathbb{P}(p_t \in \text{bin } b)$ be the bin probability. With pseudocounts,

$$\hat{\theta}_{t,b} = \frac{1 + N_{t-1,b}}{B + \sum_{j=1}^{B} N_{t-1,j}}, \qquad e_t = B\hat{\theta}_{t,b_t}, \qquad \mathbb{E}[e_t \mid \mathcal{F}_{t-1}] = B \sum_{b=1}^{B} \theta_b \hat{\theta}_{t,b}.$$

By the strong law of large numbers, $\hat{\theta}_{t,b} \to \theta_b$ almost surely, so

$$\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] \to B \sum_{b=1}^{B} \theta_b^2 \quad \text{a.s.}$$

Finally, $\sum_{b=1}^{B} \theta_b^2 \geq 1/B$, with equality iff $\theta_b = 1/B$ for all b. Since the post-shift distribution is non-uniform by assumption, the inequality is strict, so $B \sum_{b=1}^{B} \theta_b^2 > 1$. □

Proposition 3 establishes that $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] > 1$ asymptotically after a shift, under the assumption that post-change p-values are i.i.d. non-uniform. In Algorithm 1, however, the conformal p-value p_t is computed from the rank of U_t against the *entire* past sequence, which increasingly includes post-change observations. As post-change data accumulate in the reference, the rank distribution of new post-change points will migrate back toward uniformity, so the non-uniformity that drives $\mathbb{E}[e_t \mid \mathcal{F}_{t-1}] > 1$ is transient. Proposition 3 should therefore be read as characterizing the density estimator's behaviour under an exogenously supplied non-uniform p-value stream, providing a finite-horizon approximation for the regime shortly after a changepoint rather than a permanent-regime guarantee for the self-normalizing conformal construction.

We now quantify how this mean gain builds up in finite samples.

Proposition 4 (Finite-Sample Mean Gain)**.** *Let $T < t$ be a changepoint, and $n := t - T$ be the number of post-shift observations preceding time t. Define:*

- $A_b := \#\{s < T : p_s \in \text{bin } b\}$, $m := \sum_{b=1}^{B} A_b = T - 1$ *(pre-shift counts),*
- $C_b := \#\{T \leq s \leq t-1 : p_s \in \text{bin } b\}$, $\sum_{b=1}^{B} C_b = n$ *(post-shift counts),*
- $\hat{q}_b := (1 + A_b)/(B + m)$ *(smoothed pre-shift density).*

Assume post-shift p-values are i.i.d. with bin probabilities $\theta = (\theta_1, \ldots, \theta_B)$. Define

$$\gamma := B \sum_{b=1}^{B} \theta_b \hat{q}_b - 1, \qquad \delta := \sum_{b=1}^{B} \theta_b^2 - \frac{1}{B}, \tag{6}$$

where γ measures the alignment of θ with the pre-shift prior $\hat{q}$ (the factor B normalizes so that $\gamma = 0$ when $\hat{q}_b = 1/B$, i.e. when the pre-shift histogram is uninformative), and $\delta \geq 0$ measures shift intensity, vanishing iff θ is uniform. Then, conditioning on $A = (A_1, \ldots, A_B)$:

$$\mathbb{E}[e_t \mid A] = 1 + \frac{B + m}{B + m + n}\gamma + \frac{n}{B + m + n} B\delta. \tag{7}$$

Proof. The estimator and e-value are

$$\hat{\theta}_{t,b} = \frac{1 + A_b + C_b}{B + m + n}, \qquad e_t = B\hat{\theta}_{t,b_t}.$$

Since p_t independently falls in bin b with probability θ_b,

$$\mathbb{E}[e_t \mid A] = \frac{B}{B+m+n} \sum_{b=1}^{B} \theta_b \, \mathbb{E}[1 + A_b + C_b \mid A].$$

Under the i.i.d. post-shift model $\mathbb{E}[C_b \mid A] = n\theta_b$, so

$$\mathbb{E}[e_t \mid A] = \frac{B}{B+m+n} \Big[\underbrace{\sum_{b=1}^{B} \theta_b (1 + A_b)}_{(B+m)(\gamma+1)/B} + n \underbrace{\sum_{b=1}^{B} \theta_b^2}_{\delta+1/B} \Big].$$

Expanding and collecting gives (7). □

Proposition 4 shows the mean gain interpolating between two regimes: for small n the prior-alignment term γ dominates, while as $n \to \infty$ the data weight approaches one and $\mathbb{E}[e_t \mid A] \to 1 + B\delta = B \sum_b \theta_b^2$, recovering the asymptotic limit in Proposition 3. The transition speed is governed by n relative to $B + m$, with longer pre-shift histories delaying convergence.

Since γ can be negative (as low as $-m/(B+m)$ when the prior is adversarially misaligned with the shift), $\mathbb{E}[e_t \mid A]$ may initially fall below 1, meaning early e-values are expected to be less than one and dilute the e-process. From (7), the crossover $\mathbb{E}[e_t \mid A] > 1$ requires $(B+m)\gamma + Bn\delta > 0$, i.e.

$$n \;>\; n^* := \frac{(B+m)|\gamma|}{B\delta} \qquad (\gamma < 0). \tag{8}$$

This warmup length grows with the severity of misalignment $|\gamma|$ and shrinks with shift intensity δ: a larger shift overwhelms a bad prior faster. Since $|\gamma| \le m/(B+m)$, the worst-case warmup is bounded by $m/(B\delta)$, which itself shrinks as δ grows. When $\gamma \ge 0$ the prior is aligned with the shift and $\mathbb{E}[e_t \mid A] > 1$ immediately.

4.2. The Mixture E-process

A key challenge in constructing our e-process is that the true changepoint time T is unknown. An e-process starting at τ would be sensitive to drift beginning at τ but would miss earlier changes, while one starting too early would accumulate excess noise that dilutes its power. Since a weighted mixture of e-processes is itself a valid e-process by linearity of expectation (Ramdas et al., 2023), we address the unknown changepoint time by maintaining one over all possible start times:

$$M_t = \sum_{\tau=1}^{t} w_\tau \cdot M_t^{(\tau)} \tag{9}$$

where $M_t^{(\tau)} = \prod_{s=\tau}^{t} e_s$ is defined for $\tau \le t$. Since each component $M_t^{(\tau)}$ forms a valid e-process sensitive to drift beginning at τ, the mixture is sensitive to changepoints at all times. We use $w_\tau = 1/(\tau(\tau+1))$ as this sequence satisfies $\sum_{\tau=1}^{\infty} w_\tau = 1$ exactly, enabling an efficient recursion that avoids maintaining separate products for each τ:

Proposition 5 (Efficient Recursion)**.** *The mixture e-process satisfies:*

$$M_t = e_t \cdot (M_{t-1} + w_t) \tag{10}$$

Proof. Expand the definition:

$$\begin{aligned} M_t &= \sum_{\tau=1}^{t} w_\tau \cdot M_t^{(\tau)} && (11)\\ &= \sum_{\tau=1}^{t-1} w_\tau \cdot e_t \cdot M_{t-1}^{(\tau)} + w_t \cdot e_t && (12)\\ &= e_t \left(\sum_{\tau=1}^{t-1} w_\tau \cdot M_{t-1}^{(\tau)} + w_t \right) && (13)\\ &= e_t (M_{t-1} + w_t) && (14) \end{aligned}$$

□

This recursion enables a constant-time update of the mixture per observation, with additional cost from rank maintenance, while avoiding the cost of maintaining or updating all e-processes separately.

4.3. Type I Error Control

Theorem 1 (Anytime-Valid False Alarm Control)**.** *Under H_0, PITMonitor satisfies:*

$$\mathbb{P}\left(\sup_{t \geq 1} M_t \geq \frac{1}{\alpha} \right) \leq \alpha \tag{15}$$

Proof. The mixture $M_t = \sum_{\tau=1}^{t} w_\tau M_t^{(\tau)}$ is defined with $M_t^{(\tau)}$ only for $\tau \leq t$. To apply Ville's inequality, we work with an extended process defined for all $t \geq 0$. For each $\tau \geq 1$ define:

$$\widetilde{M}_t^{(\tau)} = \begin{cases} 1, & t < \tau, \\ \prod_{s=\tau}^{t} e_s, & t \geq \tau. \end{cases} \tag{16}$$

Define the full mixture over all $\tau \geq 1$:

$$\widetilde{M}_t = \sum_{\tau=1}^{\infty} w_\tau \, \widetilde{M}_t^{(\tau)}. \tag{17}$$

Since $\widetilde{M}_t^{(\tau)} = 1$ for $\tau > t$, we can expand:

$$\begin{aligned} \widetilde{M}_t &= \sum_{\tau=1}^{t} w_\tau \prod_{s=\tau}^{t} e_s + \sum_{\tau=t+1}^{\infty} w_\tau && (18)\\ &= M_t + \sum_{\tau=t+1}^{\infty} \frac{1}{\tau(\tau+1)}. && (19) \end{aligned}$$

With $w_\tau = \frac{1}{\tau(\tau+1)}$, the tail telescopes:

$$\sum_{\tau=t+1}^{\infty} \frac{1}{\tau(\tau+1)} = \sum_{\tau=t+1}^{\infty} \left(\frac{1}{\tau} - \frac{1}{\tau+1} \right) = \frac{1}{t+1}.$$

Hence

$$M_t \leq M_t + \frac{1}{t+1} = \widetilde{M}_t, \tag{20}$$

$$\therefore \left\{ \sup_{t \geq 1} M_t \geq \frac{1}{\alpha} \right\} \subseteq \left\{ \sup_{t \geq 1} \widetilde{M}_t \geq \frac{1}{\alpha} \right\}. \tag{21}$$

Since each $(\widetilde{M}_t^{(\tau)})_{t \geq 0}$ is a non-negative supermartingale with $\widetilde{M}_0^{(\tau)} = 1$. By Tonelli's theorem,

$$\mathbb{E}[\widetilde{M}_t \mid \mathcal{F}_{t-1}] = \sum_{\tau=1}^{\infty} w_\tau \, \mathbb{E}[\widetilde{M}_t^{(\tau)} \mid \mathcal{F}_{t-1}] \tag{22}$$

$$\leq \sum_{\tau=1}^{\infty} w_\tau \, \widetilde{M}_{t-1}^{(\tau)} = \widetilde{M}_{t-1}, \tag{23}$$

so $(\widetilde{M}_t)$ is a non-negative supermartingale with $\widetilde{M}_0 = \sum_{\tau=1}^{\infty} w_\tau = 1$. Finally, applying Ville's inequality,

$$\mathbb{P}\left(\sup_{t \geq 1} M_t \geq \frac{1}{\alpha} \right) \leq \mathbb{P}\left(\sup_{t \geq 1} \widetilde{M}_t \geq \frac{1}{\alpha} \right) \leq \alpha. \tag{24}$$

□

4.4. Changepoint Estimation

After an alarm at time T, we estimate the changepoint location by selecting the split that best explains the post-split p-values as non-uniform. For each candidate $s \in \{1, \ldots, T-1\}$, we compute a split-specific histogram $\mathbf{n}^{(s)} = (n_1^{(s)}, \ldots, n_B^{(s)})$, where $n_b^{(s)}$ counts p-values in bin b and $N^{(s)} = \sum_b n_b^{(s)} = T - s$, and evaluate the segment $(p_{s+1}, \ldots, p_T)$ under two hypotheses:

- $\mathbf{H_0^{(s)}}$: p-values are Uniform$(0, 1)$, so each of B bins receives probability $1/B$.
- $\mathbf{H_1^{(s)}}$: bin probabilities are unknown, with a symmetric Dirichlet prior Dir$(\kappa, \ldots, \kappa)$.[4]

We use a Bayes factor rather than a likelihood ratio because H_1 would otherwise trivially outperform H_0 by overfitting, since it contains H_0 as a special case. Averaging over the Dirichlet prior penalizes H_1 for the probability mass it wastes on configurations the data does not support. We set $\kappa = 1/2$, since it is invariant to the reparametrization of bin boundaries (Jeffreys, 1961).

Under H_0 every bin has probability $1/B$, so the multinomial log-likelihood reduces to:

$$\log p(\mathbf{n}^{(s)} \mid H_0^{(s)}) = \log \frac{N^{(s)}!}{\prod_b n_b^{(s)}!} - N^{(s)} \log B \tag{25}$$

[4] It should be noted that this is a heuristic approximation since the p_t sequence carries dependence across time.

Algorithm 1 PITMonitor

Require: Significance level α, number of bins B

Initialize: $M_0 \leftarrow 0$, histogram counts $c_1, \ldots, c_B \leftarrow 1$
for $t = 1, 2, \ldots$ **do**
  Observe PIT $U_t \in [0, 1]$
  Insert U_t into sorted list; compute rank R_t
  Sample $V_t \sim \text{Uniform}(0, 1)$
  $p_t \leftarrow (R_t - 1 + V_t)/t$ ▹ Conformal p-value
  $b \leftarrow \min(\lfloor p_t \cdot B \rfloor + 1,\ B)$ ▹ Integer bin count
  $e_t \leftarrow B \cdot c_b / \sum_{j=1}^{B} c_j$ ▹ E-value from density
  $c_b \leftarrow c_b + 1$ ▹ Histogram update
  $w_t \leftarrow 1/(t \cdot (t+1))$ ▹ Mixture weight
  $M_t \leftarrow e_t \cdot (M_{t-1} + w_t)$ ▹ Mixture e-process
  if $M_t \geq 1/\alpha$ **then** ▹ Threshold
    return ALARM at time t
  end if
end for

Under H_1, the probabilities are unknown so we integrate them out over the Dirichlet prior:

$$\begin{aligned}\log p(\mathbf{n}^{(s)} \mid H_1^{(s)}) = \log \frac{N^{(s)}!}{\prod_b n_b^{(s)}!} + \log \Gamma(B\kappa) - \log \Gamma(N^{(s)} + B\kappa) \\ + \sum_{b=1}^{B} \left[\log \Gamma(n_b^{(s)} + \kappa) - \log \Gamma(\kappa)\right]\end{aligned} \tag{26}$$

Since the combinatorial factor appears in both likelihoods, it cancels in the log Bayes factor:

$$\begin{aligned}\log \mathrm{BF}(s) &= \log p(\mathbf{n}^{(s)} \mid H_1^{(s)}) - \log p(\mathbf{n}^{(s)} \mid H_0^{(s)}) \\ &= \log \Gamma(B\kappa) - \log \Gamma(N^{(s)} + B\kappa) \\ &\quad + \sum_{b=1}^{B} \left[\log \Gamma(n_b^{(s)} + \kappa) - \log \Gamma(\kappa)\right] \\ &\quad + N^{(s)} \log B\end{aligned} \tag{27}$$

We then identify the changepoint simply as $\hat{\tau} = \arg\max_s \log \mathrm{BF}(s)$.[5]

This localization capability is absent from all `river` baselines, which expose only a binary alarm flag. PITMonitor thus enables practitioners not just to detect that drift has occurred, but to identify approximately how far back model outputs may have been affected by the shift - directly actionable for deciding how much historical inference to distrust or recompute.

4.5. Complete Algorithm

5. Experiments

We evaluate PITMonitor on the FriedmanDrift benchmark, a standard regression task for evaluating concept drift detectors under controlled conditions, comparing against the seven

[5] A natural alternative is the start time of the largest e-process, $\arg\max_\tau M_T^{(\tau)}$, but following from Proposition 4, we expect $e_s \approx 1$ for some time after a changepoint, biasing this alternative estimator toward later start times.

stream-drift detectors included in `river`. Although FriedmanDrift induces covariate drift, feeding shifted inputs to a held-fixed model changes the PIT distribution, so covariate drift directly becomes calibration drift from the monitor's perspective.

5.1. Setup

Dataset and drift scenarios. FriedmanDrift (Montiel et al., 2021) is a synthetic regression stream with 10 input features (x_0–x_9). We evaluate three drift types that represent qualitatively different distribution changes:

- **GRA** (Global Recurring Abrupt): All features change simultaneously at an abrupt onset, representing a rapid regime shift.
- **GSG** (Global Slow Gradual): The change spreads linearly across all features over a 500-sample transition window, representing gradual covariate shift.
- **LEA** (Local Expanding Abrupt): Drift starts on a subset of features and expands to include more over time, representing localized distribution change.

Stream layout. A training stream consisting of $n_{\text{train}} = 10{,}000$ pre-drift samples is generated, along with 10,000 independent test streams containing $n_{\text{stable}} = 2{,}500$ pre-drift monitoring samples that define the null-hypothesis window for FPR estimation, and $n_{\text{post}} = 2{,}500$ post-drift samples for TPR estimation. The drift onset occurs at the boundary between the pre- and post-drift segments.

Predictive model. We train a feedforward neural network outputting a Gaussian predictive distribution $\mathcal{N}(\mu_t, \sigma_t^2)$ for each input. The network has 3 hidden layers of 128 units with SiLU activations. Inputs and targets are standardized using per-feature means and standard deviations fitted on the training set. Training uses mini-batches of size 256, the Adam optimizer with initial learning rate 3×10^{-4}, a cosine annealing learning rate schedule, and 500 epochs. The model achieves $R^2 = 0.96$ on a held-out pre-drift test set, with an expected calibration error (ECE) of 0.01, confirming it is well-specified and calibrated before monitoring begins. The model is trained once and subsequently held fixed across all 10,000 Monte Carlo trials.

PIT construction. For each monitoring sample (x_t, y_t) the PIT is:

$$U_t = \Phi\left(\frac{y_t - \mu_t}{\sigma_t}\right) \tag{28}$$

where μ_t, σ_t are the predicted mean and standard deviation and Φ is the standard normal CDF.

Detector settings. All `river` baselines are run with their library-default parameters. This reflects an out-of-the-box comparison, avoids detector-specific tuning in favour of any particular method, and sidesteps the need for held-out null data or advance knowledge of the monitoring window length to align parameters with a target stream-level FPR.

- **PITMonitor**: Significance level $\alpha = 0.05$, number of bins $B = 100$.
- **Continuous-input baselines (ADWIN, KSWIN, PageHinkley)**: library-default parameters.

Table 1
Drift detection results on FriedmanDrift (10,000 trials, $\alpha = 0.05$).

	GRA			GSG			LEA		
Method	TPR	FPR	Delay	TPR	FPR	Delay	TPR	FPR	Delay
PITMonitor	96.2%	3.8%	77	96.2%	3.8%	189	96.2%	3.8%	1919
ADWIN	99.1%	0.9%	27	99.1%	0.9%	27	99.1%	0.9%	115
KSWIN	2.9%	97.1%	16	2.8%	97.2%	172	2.8%	97.2%	656
PageHinkley	0.4%	99.6%	1	0.4%	99.6%	6	0.4%	99.6%	70
DDM	90.8%	9.2%	405	90.0%	9.2%	666	22.0%	9.2%	2050
EDDM	8.7%	91.3%	344	8.7%	91.3%	562	1.6%	91.3%	1759
HDDM_A	94.0%	6.0%	60	94.0%	6.0%	168	52.5%	6.0%	2032
HDDM_W	9.4%	90.5%	15	9.4%	90.5%	46	9.4%	90.5%	586

- **Binary-input baselines (DDM, EDDM, HDDM_A, HDDM_W)**: library-default parameters. These detectors require a binary error signal; we binarize via $b_t = \mathbf{1}[|r_t| > \theta]$ where θ is the *median* of $|r_t|$ on the training data. Since these methods were designed for classification, no guidance exists for thresholding, and the median is the most assumption-free choice.

Evaluation protocol. We run $N = 10{,}000$ Monte Carlo trials per scenario, each using a distinct random seed. For each trial we record whether an alarm fires, its index, and whether it occurred before or after the true drift onset, as well as distance from the true changepoint for PITMonitor. We report:

- **TPR**: fraction of trials with a true-positive alarm (alarm fired after drift onset).
- **FPR**: fraction of trials with a false alarm (alarm fired before drift onset).
- **Mean detection delay**: mean number of samples between the true drift onset and the alarm, over true-positive trials only.

5.2. Results

Table 1 presents the full results. Figure 1 visualizes TPR and FPR per method and scenario, Figure 2 shows a representative single-run monitoring trace for the GSG scenario, and Figure 3 shows the distribution of detection delays over true-positive trials. The null window is drawn from the same pre-drift distribution for all three scenarios, so each detector's FPR is near-identical across scenarios by construction.

Type I error control. PITMonitor achieves FPR of 3.8% across all scenarios, consistent with the nominal $\alpha = 0.05$ guarantee in Theorem 1. In this benchmark, ADWIN yields 0.9% empirical FPR on the 2,500-sample null window, but this number is specific to the chosen horizon and dataset. Among the other baselines, DDM and HDDM_A achieve 9.2% and 6.0% FPR, while KSWIN, PageHinkley, EDDM, and HDDM_W exceed 89% FPR with default settings; at these levels, high TPR is a trivial consequence of near-constant alarming rather than detection power, so their TPR and delay figures are not meaningfully comparable. This empirical FPR comparison should be read with care: FriedmanDrift is stationary before the changepoint by construction, so it does not stress cases with persistent low-level null violations (e.g., weak temporal dependence).

Global recurring abrupt drift (GRA). On GRA, PITMonitor achieves 96.2% TPR with 3.8% FPR and mean delay 77 samples. ADWIN attains higher TPR (99.1%) and faster delay (27), but its false-alarm behavior here remains an empirical estimate tied to this finite monitoring window. DDM and HDDM_A reach 90.8% and 94.0% TPR respectively.

Global slow gradual drift (GSG). On GSG, PITMonitor achieves 96.2% TPR with 3.8% FPR and mean delay 189 samples. ADWIN again records higher TPR (99.1%) and faster delay (27), while DDM and HDDM_A reach 90.0% and 94.0% TPR respectively.

Local expanding drift (LEA). On LEA, PITMonitor maintains 96.2% TPR with 3.8% FPR, but mean delay rises to 1919 samples. This pattern is consistent with the expanding-drift structure: early phases induce weaker PIT distortion, so evidence accumulates slowly until later expansion stages. ADWIN detects earlier in this scenario (115 mean delay, 99.1% TPR), indicating a speed-guarantee tradeoff in this benchmark.

Detection delays. Figure 3 reports delay distributions over true-positive trials only. Accordingly, it should be interpreted jointly with TPR; a detector with low TPR may have a tight delay distribution simply because it detects only the easiest cases. PITMonitor's delay distribution is tightest under GRA, where the shift is strongest and most immediate, and widest under LEA, where the shift is weakest at onset and evidence accumulates slowly. The other detectors show similar patterns, with ADWIN achieving the fastest detection among detectors with a TPR > 90%.

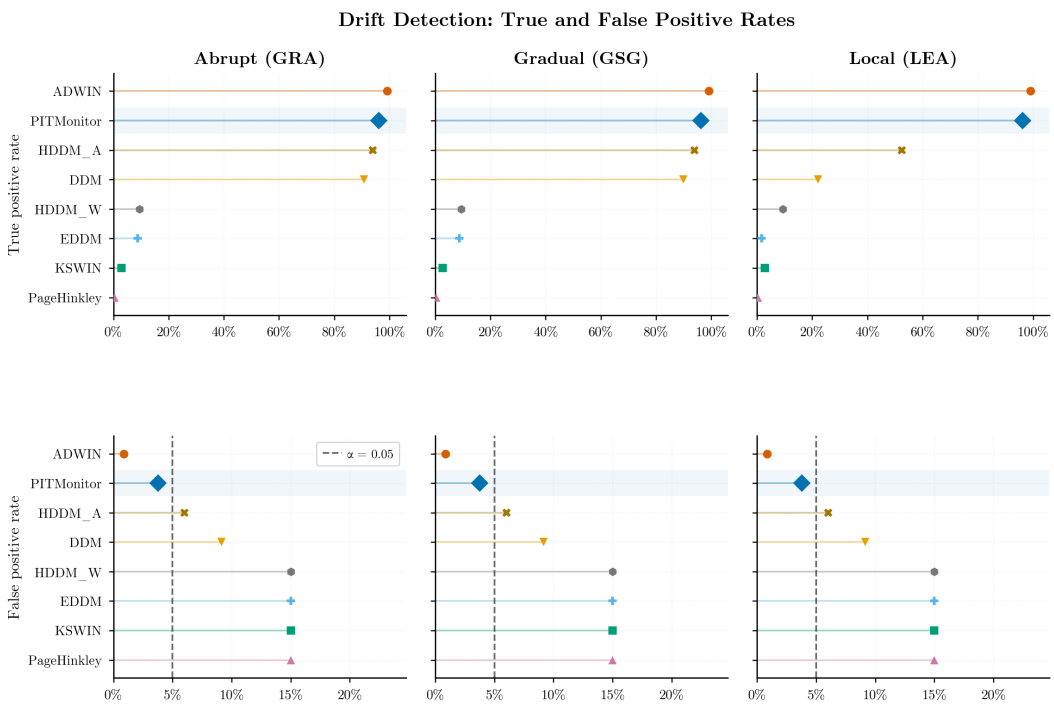

Fig 1. *TPR and FPR across all detectors and drift scenarios. The dashed line marks* $\alpha = 0.05$.

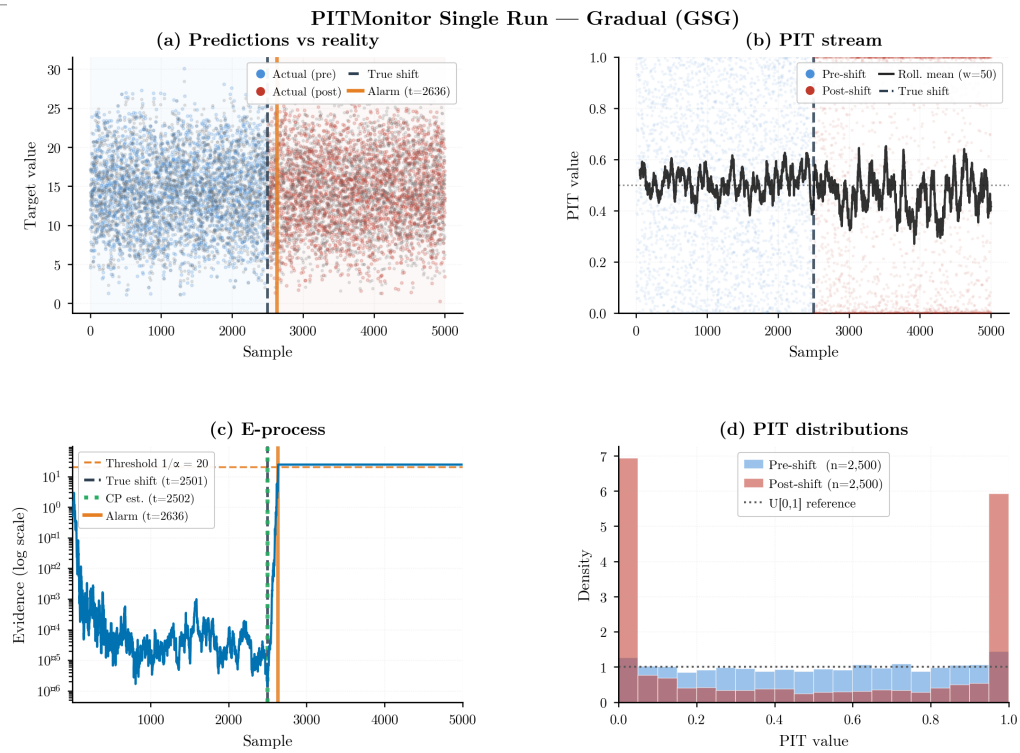

Fig 2. *Single-run PITMonitor trace. (a) Predicted vs. actual values. (b) PIT stream with rolling mean. (c) Mixture e-process on a log scale; dashed line at the threshold* $1/\alpha$. *(d) Pre- and post-shift PIT histograms.*

Changepoint localization. On GRA and GSG, mean absolute error is 1.2 and 6.8 samples, indicating accurate localization under global drift. On LEA, mean error increases to 1839.3. This is expected because LEA induces multiple transition points during expansion, violating our single-changepoint localization assumption; the estimator therefore tends to select the split with strongest post-split non-uniformity, which typically occurs in later expansion phases (near offset 1,666) rather than at initial onset. In this regime, MAE should therefore be interpreted as a mismatch to a single-reference onset, not necessarily as failure to identify the most operationally relevant transition phase.

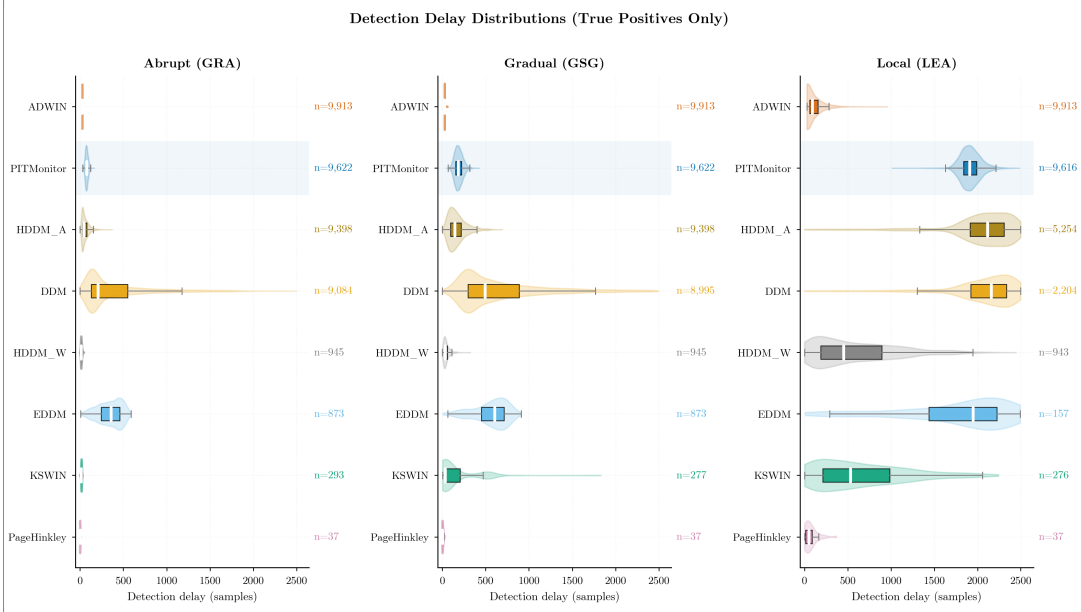

Fig 3. *Detection delay distributions (true-positive trials only) across all detectors and drift scenarios. Violins show the full delay distribution, while the white bar marks the median.*

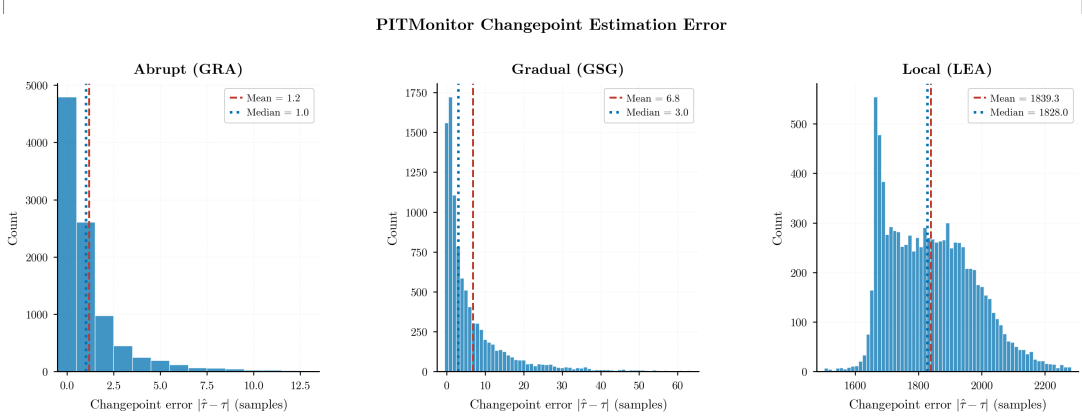

Fig 4. *Distribution of PITMonitor changepoint estimation error $|\hat{\tau} - \tau|$ across true-positive trials.*

6. Discussion

Scope. PITMonitor is intended for continuous monitoring of deployed probabilistic models when false alarms are costly, monitoring duration is open-ended, and calibration drift (not only accuracy drift) is operationally important. For one-time calibration checks or accuracy-only monitoring without need of false-alarm control or changepoint localization, standard diagnostics remain appropriate.

Limitations. *Expanding drift.* PITMonitor detects LEA drift with 96.2% TPR but with 1919 mean delay, reflecting slower evidence accumulation when only a subset of features has shifted.

Calibration improvements. PITMonitor detects any departure from exchangeability in the PIT sequence and does not distinguish between deterioration and improvement in calibration. A model that becomes better calibrated mid-stream may also trigger an alarm. In practice, the direction of drift can be assessed by inspecting the PIT histogram following an alarm.

Changepoint localization on expanding drift. Localization is less reliable under sequential expansion: the estimator targets the phase of strongest distributional change, which can lag the true onset. This interpretation is supported by additional multi-step localization checks reported in Appendix A.

Detection delay vs. FPR control. Anytime-valid control can increase delay relative to methods without stream-level guarantees. In practice, α should be set against long-run false-alarm tolerance rather than treated as a fixed convention.

Exchangeability assumption. PITMonitor tests exchangeability of PITs. Non-exchangeability can also arise from temporal dependence: autocorrelated PITs can trigger alarms even when the calibration distribution is unchanged. This has two implications. First, the i.i.d.-based power statements in Section 4.1 (including Proposition 4) are clean reference results, not finite-sample guarantees for dependent streams. Second, mild but persistent null violations can in principle inflate long-run alarms. In our FriedmanDrift experiments this did not produce extreme empirical FPR, but the benchmark is stationary pre-drift by design, so this robustness requires broader evaluation on dependent and non-stationary streams. Practitioners should check autocorrelation in PITs (e.g., Ljung-Box or ACF diagnostics); a practical mitigation is thinning the PIT sequence, trading reduced dependence for slower detection. Appendix A summarizes auxiliary dependent/non-stationary stress tests used to inform this discussion.

Persistence requirement. PITMonitor reacts to accumulated evidence, not single-step anomalies. Under non-exchangeability, the adaptive histogram attempts to track the current concentration pattern of p-values; when tracking is accurate, expected log-growth is positive (i.e., $\mathbb{E}[\log e_t] > 0$) and cumulative evidence increases until crossing the alarm threshold. If deviations are brief, rapidly reversing, or drift faster than the estimator can track, growth is reduced or intermittent, so alarms may be delayed or absent.

Practical recommendations. *Histogram resolution.* The number of bins B controls the bias-variance tradeoff in the density estimator. Smaller B is more stable but slower to adapt; larger B adapts faster at the cost of more variance in the estimated density and should be scaled with the expected number of monitoring samples. Additionally, during the early monitoring

phase (roughly the first B observations), the histogram is dominated by the pseudocounts and e-values are nearly 1 regardless of the data, yielding low sensitivity. Practitioners expecting drift immediately after deployment may wish to use a smaller B or include a warm-start period before monitoring begins.

Structural comparison to `river` baselines. *Parameter simplicity and FPR guarantees.* In contrast to PITMonitor's two parameters (α bounding stream-level FPR, B setting histogram resolution), the `river` baselines expose many internal sensitivity parameters whose relationship to stream-level FPR is indirect, horizon-dependent, and formally unguaranteed. Aligning these methods to a target stream-level FPR typically requires advance knowledge of the monitoring window or empirical calibration.

Calibration specificity. PITMonitor operates on the full PIT distribution and therefore has the capacity to detect shifts that change distributional moments (location, scale, skewness, and beyond), insofar as these induce non-exchangeability in the PIT stream. In contrast, many `river` baselines are primarily tuned to mean-like shifts in monitored signals.

Benchmark context. FriedmanDrift is `river`'s own canonical synthetic benchmark; the strong performance of library-default parameters on this benchmark should be interpreted in that context.

7. Conclusion

We presented PITMonitor, an anytime-valid method for monitoring calibration of deployed probabilistic models. By testing exchangeability of probability integral transforms using a mixture e-process, PITMonitor enables continuous monitoring without inflating Type I error, regardless of when or why monitoring stops.

Experiments on three FriedmanDrift scenarios, with baselines at library defaults, show that PITMonitor attains competitive TPR on global drift (96.2% vs. ADWIN's 99.1%) while keeping empirical FPR under the target level (3.8% at $\alpha = 0.05$). On local expanding drift, PITMonitor retains comparable TPR but with substantially larger delay, reflecting the slower evidence accumulation when only a subset of features has shifted.

Three structural advantages persist beyond the numerical comparison. First, PITMonitor's FPR guarantee is horizon-independent: PITMonitor provably satisfies $\mathbb{P}(\text{ever alarm} \mid H_0) \leq \alpha$ regardless of deployment duration. Second, PITMonitor targets calibration specifically: by operating on probability integral transforms rather than residuals or error rates, it is sensitive to changes in predictive uncertainty that leave accuracy metrics unchanged and would be missed by residual-based detectors. Third, it provides actionable changepoint estimates that identify how far back model outputs were corrupted, directly informing decisions about how much historical inference to distrust or recompute.

In practice, PITMonitor is most suitable when formal long-run error control, changepoint localization, and calibration-specific monitoring are priorities. When gradually expanding drift is anticipated, combining PITMonitor with a faster residual-based detector can improve early warning at the cost of weaker long-run false-alarm guarantees.

Future work has four directions. First, for temporally dependent predictions, we aim to develop a dependence-robust construction (beyond pragmatic thinning, which reduces power). Second, for multivariate outputs, a practical baseline is to reduce each prediction-outcome

pair to a scalar PIT and reuse the same monitoring pipeline; developing fully multivariate transforms (e.g., Rosenblatt transforms) that better preserve joint dependence remains open. Third, we aim to improve detection power and changepoint localization under partial distributional shifts. Fourth, we aim to couple detection with automatic post-alarm recalibration: after an alarm, estimate the post-alarm empirical PIT density $\hat{q}(z)$ and reweight the predictive density as $\hat{f}(y) = p(y)\,\hat{q}(z)^{\lambda(e)}$, where $\lambda(e)$ is a shrinkage weight that decreases with uncertainty and is learned from e-process evidence. This would tie correction strength directly to detected miscalibration severity, yielding conservative updates under weak evidence and stronger recalibration when evidence accumulates.

Code Availability. https://github.com/tristan-farran/pitmon.

Appendix A: Additional Validation Checks

To keep the main text focused on the core benchmark and theorem-driven claims, we report supplementary checks here. These diagnostics are not used for model or detector tuning and are not part of the primary comparative evaluation; they are included to document robustness and interpretation boundaries.

Finite-time proposition sanity check. The finite-sample mean formula from Proposition 4 is numerically consistent with Monte Carlo estimates for representative aligned and misaligned priors, with maximum absolute discrepancies of 0.004 (aligned prior) and 0.004 (misaligned prior). This check is a numerical consistency test of the proved expression rather than independent additional evidence.

Dependent and non-stationary PIT stress tests. Using $\alpha = 0.05$, $B = 100$, and $n = 2{,}000$ trials per regime, the i.i.d. uniform baseline yields an alarm rate of 4.0% (95% CI [3.2%, 5.0%], median alarm time 50). Under a stationary dependent AR(1)-copula stream with uniform margins, the alarm rate increases to 11.5% (95% CI [10.2%, 13.0%], median alarm time 54), supporting the temporal-dependence caveat in the Limitations paragraph. For comparison, mild non-stationarity gives 3.5% (95% CI [2.8%, 4.5%], median alarm time 42), and seasonality gives 5.0% (95% CI [4.1%, 6.0%], median alarm time 52).

Multi-step localization interpretation check. In a staged step-drift stream, changepoint localization is substantially closer to the maximal-intensity phase than to initial onset: MAE to onset is 1814.1 versus 265.1 to the strongest phase, and 1,926 of 2,000 runs are closer to the strongest phase reference. This result is used to interpret localization behavior under expanding/staged drift, not as a primary benchmark metric.

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